

Minutes of CFS++ Developer's Meeting

Participants: Bahr, Escobar(minutes responsible), Hauck, Link

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1 Situation of current activities

- *GiD as pre- postprocessor tool for CFS++:* (Andreas, Martin) Andreas and Martin have tested successfully the external software package GiD (the personal pre and post processor) version 7.7.0b as a tool for the mesh generation and visualization of results generated by CFS++. Additional proprietary features were included inside GiD which makes it more compatible with the CFS++ mesh generation standards and which allow reproducibility. These features are under the menu options **Regions**, **Scripting** and **Mesh**. GiD will be the default pre- and postprocessor tool for the course CAESAR during the winter term. Further GiD documentation is found in the CFS++ manual. Unfortunately, GiD is still significantly slower for large (mainly 3D) problems in comparison to other postprocessor tools (e.g. GMV).
- *Performance Suit:* (Ludwig, Gerhard, Max) Ludwig and Gerhard have created and tested a first version of a performance suit. This implementation allows the evaluation of time and memory consumption in CFS++ for the different solvers available. The main objective here is to evaluate the performance of CFS++ for middle and large problems. The performance suit generates as result **.profile** files with different suffixes corresponding to the solvers being tested. In the future the PERFORMANCESUIT should work in an analogous way to the TESTSUITE®. At the moment, since the main script is written in AWK, the changing of directories seems to be AWKward, therefore Ludwig and Max will adapt it to Perl.
- *Improvement to testsuit:* Andreas has proposed a small improvement in the testsuit in the case it does not find differences between the new results and the reference ones. For this case it should write a message **“REFERENCE RESULTS AND NEW RESULTS MATCH”** instead of the current **“NO DIFFERENCES FOUND”** since this would provide better visibility when screening the **.log** file.
- *Implementation of new preconditioners:* (Manfred) Manfred has implemented an Incomplete Choleski (**IC0**) preconditioner which for large problems allows a reduction in memory consumption of about 45% in comparison to the **ILU0** preconditioner. Concerning time performance, currently, due to a simplification in the implementation, it needs about 30% more iterations than **ILU0**, which makes it slower. A progress report during the factorization for this preconditioner is also included in the **.las** file. An implementation of an **ILDL0** is also planned.
- *Non-matching grids:* (Simon Triebenbacher) As topic for his Master's thesis, Simon Triebenbacher is working on a CVS project branch in the implementation of a non-matching grid capability for CFS++. His work should be ready in January which means that after that date, it could be consider for inclusion in the CFS++ project's trunk.
- *Restart file:* (Gerhard) A new possibility for resuming a CFS++ mechanical simulation from a specific time step has been derived from the memento implementation. A new

XML element has been added in the transient section of the **.XML** file. If the option is activated, CFS++ generates a restart file at the specified restart increments and by default at the last time step of the computation. The restart file has the same name as the problem name and a suffix `_[pdename].restart`. To read the restart file, the command line option `-r` has been added. Further details in the CFS++ manual. No significant overhead is added to the computation apart from the writing of the restart file at the specific time step. Andreas will adapt the WriteHistory function so that the results after the restart are appended to the old history files. As mentioned above, at the moment it works only for the mechanical field but its extension to other physical fields should not pose major problems. Immediate extension for acoustics will be done by Max.

- *Pressure from acoustic Potential + Inhom. Neumann BC in Acoustics:* (Ludwig) There is just a little bug in the Neumann BCs to be solved. A point to discuss in the future is the fact, that all integrators in the acoustic PDE are multiplied with density because of coupling to mechanical PDE. One could think about multiplying the integrators with density only when coupling to other PDEs?

2 Documentation in the xmlReference file

Always a touchy issue in meetings... this time regarding the documentation of the XML parameter definitions

- *Gerhard:* will document mechanics
- *Ludwig:* will do acoustics
- *Max:* will do MpCCI
- *Andreas:* will provide documentation to direct coupling, local coordinate systems ...
- orphans fields looking for developers to adopt (document) them: piezo, material parameter identification...

3 Main tasks

- *Gerhard:* will try to tackle nonlinear mechanics in respect to material description and contact mechanics
- *Ludwig:* tries to implement heat conduction (will be heatCondPDE.cc and .hh soon.) and coupling to acoustics, i.e. a rhs source vector from acoustics alters heat conduction PDE
- *Manfred:* new material file format and file reader, spectral elements in time domain, PML, an explicit solver, Freibier and all the good stuff
- *Max:* restart for acoustics, continuing with the higher order absorbing BCs
- *Andreas:* enhances WriteResults and StoreSolution class, split off history file handling, sophisticated shell elements